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ORIGINAL RESEARCH

Heuristic Segmentation Assisted Deep-Spatial Feature Learning Model for Leprosy Detection

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Abstract

The high pace of rising leprosy cases and resulting socio-stigma has alarmed academia-industries to achieve non-invasive, touchless, and vision-based computer aided diagnosis (CAD) solution for leprosy detection and classification. Unlike classical CAD systems, leprosy detection and classification has remained a least explored research area. Invasive and chemical assay-based leprosy detection approaches are often found time-consuming, complex, and limited to cope-up with real-world's demand. Though, in the last few years every effort has been made towards vision-based leprosy detection and classification; however, limited data, inferior feature, high-annotation demands, and importantly low accuracy limit their suitability. Considering these facts as motivation, in this paper a highly robust Heuristic Driven Segmentation assisted hybrid deep-spatio-textural feature learning for leprosy detection and classification is developed. It emphasizes on both region of interest segmentation and its optimization, as well as feature improvement to achieve higher accuracy and reliability. Here, firefly heuristic driven Fuzzy C-Means clustering (FFCM) was developed to perform ROI specific segmentation, which was followed by a hybrid deep spatio-textural feature extraction process. Noticeably, FFCM at one hand enabled automatic and accurate ROI-segmentation. On the contrary, the use of hybrid features including descriptive spatio-textural textural statistics (i.e., Gray-level co-occurrence metrics) and 4096-dimensional AlexNet features provided an intrinsic feature rich vector to perform accurate leprosy classification. The ROI-specific hybrid features (i.e., GLCM and AlexNet features) were processed for two-class classification using ensemble random forest classifier that labelled each input skin lesion as the normal image or the leprosy detected. The proposed leprosy detection method exhibited superior performance (accuracy = 96.6%, precision = 99.7%, recall = 95.8%, F-score = 0.9771) over other state-of-art methods like GLCM (accuracy = 88.51%) and Convolutional Neural Network based methods (91%). It affirms its suitability of the proposed CAD model towards automated and touchless real-time leprosy diagnosis.

Keywords Leprosy detection and classification · Heuristic-based lesion segmentation · Hybrid spatio-textural deep features · GLCM · AlexNet · Ensemble classification

Introduction

Leprosy or the Hansen's disease represent a chronic infectious disease, which is primarily caused by Mycobacterium leprae. Typically, leprosy is characterized by immunological reactions, often triggered by the human body's immune response against Mycobacterium leprae [1]. Such kinds of human body reactions can take place before, during or after treatment with multi-drug therapy (MDT) [1, 2]. If unrecognized or improperly managed, human body reactions can turn into irreversible nerve damage and deformities. The impact of leprosy can be found in the vicinity of the eyes, nerves and the mucosal surfaces pertaining to the upper respiratory tract [2]. Although leprosy is a curable disease, delay in

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(early) detection and allied diagnosis can prolong it to years, thus impacting the health, social and mental well-being of an individual [2, 3]. The delayed diagnosis can also give rise to different health adversaries or disabilities impacting overall well-being. Leprosy being a communicable disease spreads amongst human by means of droplets escaped out of nose and mouth. Consequently, it increasesdisease cases to the one with frequent and the close contacts [2, 3]. The high pace rising cases of leprosy, especially in India, Brazil, Indonesia, Nepal, Congo, Somalia etc.,has alarmed global healthcare agencies or organizations to recognize leprosy disease as the Neglected tropical diseases (NTDs) [4]. The severity of this disease can be understood by the fact that it has been declared as a communicable disease by WHO and has been prevalently found in tropical and subtropical conditions in 149 countries globally. The impact of this disease can be found across an estimated population of more than one billion globally, despite making billions of investments every year to cure it and suppress the rise in cases. Recent studies reveal that the lack of sanitation, unhygienic living conditions, especially over crowded human habitants might get infected easily after getting in contact of an infectious vector. Unfortunately, India contributes to nearly 57.7%of the global burden in leprosy related cases (India Leprosy Data, 2018) [5–7].

Unlike other skin diseases like skin cancer melanoma, leprosy is often found to be non-linear in structure making diagnosis more complex and false-positive prone [8]. Moreover, the stereotype approaches are more complex and time-consuming that suppresses its scalability to cope up up-surging demands [9]. Furthermore, it requires skilled resources and experts that not only makes it human dependent but time-consuming [10]. The delayed treatment and resulting complexities turn into a socially neglected and hidden disease that becomes severe and disastrous in later period [11, 12]. Though, to detect leprosy especially in the regions with low lab facility and to reduce invasive assessment, WHO recommends leprosy profiling based on the lesions available on skin [12]. This approach is based on a hypothesis that a skin lesion with five or less lesions with no evident or quantifiable bacilli on skin smears, would be classified as the paucibacillary (PB) leprosy. On the contrary, a lesion with only one lesion, is labelled as a single lesion PB. On the other hand, more (i.e., > six) lesions possessing positive skin smear test can be labeled as multibacillary (MB) kind of the leprosy [1–5]. However, such assessment requires very experienced doctor or healthcare professional to examine skin lesions and a mistake in accurate assessment often leads wrong diagnosis giving rise to the more severe health complications. Furthermore, it requires manual skin-assessment which is quite complex and difficult to generalize in current social conditions and hence limits such approaches to meet the demand. Though, some other physical behaviors of malfunctions like sensory loss with typical red-skin patches, numbness in extremities especially with paresthesia, painless burns, earlobes swelling, swelling in lumps etc. indicate symptoms of the leprosy [5, 6]., Despite suitability the lack of data, annotations and standard technologies makes this approach susceptible and false-prone [13]. Unlike invasive and hypothetical approaches of leprosy detection, vision-based approaches can yield a more efficient solution. The ability to exploit spatial and textural features of the region of interest (i.e., leprosy lesions) can enable vision-based methods performing real-time leprosy detection and allied diagnosis. The spatio-textural characteristics such as non-linear patterns (i.e., change in color, contrast, dimensions, etc.) can be applied to perform leprosy detection [14]. To achieve this a few efforts have been made by performing two-dimensional image analysis and image processing. Vision-based methods can help in learning the skin lesions for further pattern classification and hence leprosy detection; yet distinguishing ROI from the insignificant or ambiguous backgrounds (say, neighboring pixel elements), feature extraction over unannotated lesions and classifying them optimally has remained a challenge [14]. In fact, vision-based computer aided diagnosis (especially 2D image analysis-based disease detection and classification) requires CAD to learn the optimal set of ROI-sensitive features and yield highly accurate classification. Therefore, it requires the overall system to have the robust feature extraction and learning tools to yield accurate leprosy detection solution [15]. A few efforts have been made where authors have mainly focused on exploiting local spatio-textural features to perform leprosy detection and classification. Some of the existing methods have employed local features like GLCM, local binary patterns (LBP), discrete cosine coefficients, wavelet features etc. [16] to learn and detect leprosy in provided skin lesions. However, non-linearity in skin lesions, minimally annotated data and diversity of intrinsic lesion patterns imposes false positive, thus making them low accurate models [16, 17]. The key reason behind such inferiorly performing models can be ambiguous ROI definition and inferior feature vector [15]. In the past, a few methods were proposed by applying deep learning concepts like convolutional neural network (CNN) [27], transferable deep learning methods [18–22], etc.; however, most of these existing methods avoid such complexities (i.e., non-linear patterns, ROI-independent cumulative feature learning and shallow local feature learning, etc.) [23–25]. The depth assessment reveal that applying local spatio-textural features along with high-dimensional deep feature learning can enable superior learning and hence higher accuracy. Moreover, performing ROI-sensitive feature extraction and

learning can help reducing insignificant feature learning and hence can suppress false-positive performance [16, 17].

The proposed model focused on optimal ROI-segmentation followed by ROI-specific hybrid feature extraction and learning to achieve higher accuracy and reliability. In the proposed model, at first input leprosy skin lesions are processed for pre-processing tasks followed by a robust evolutionary computing driven Fuzzy C-Means clustering to segment (leprosy-specific) lesions. In the proposed method, a heuristic method named the Firefly algorithm is applied in sync with FCM to segment leprosy's ROI. Once segmenting the ROI, it executes a hybrid feature extraction method by applying GLCM and AlexNet-CNN individually. It extracts statistically diverse and information rich GLCM features like Contrast, Homogeneity, Energy, Entropy and Correlation. Similarly, a well-known transferable deep learning method named AlexNet-CNN was applied to extract 4096 dimensional deep features over the segmented ROI. Subsequently, the proposed method amalgamates both GLCM and AlexNet Features to retrieve a hybrid feature vector which is later learnt using random forest ensemble learning method to annotate each input lesion images as leprosy or normal lesion classes. The simulation results over the different sets of input test cases revealed that the proposed hybrid feature driven leprosy detection system achieves the classification accuracy of 96.6%, while maintaining average sensitivity and specificity of 95.7% and 95.9%, respectively. The relative performance assessment confirms robustness of the proposed hybrid feature based leprosy detection solution over standalone GLCM or AlexNet feature based models. Moreover, depth performance assessment confirmed superiority of the proposed hybrid feature driven model in comparison to the existing approaches. The functional as well as performance robustness confirms its suitability towards real-time leprosy detection and swift diagnosis purposes.

This manuscript is divided as follows. Section II discusses the related works pertaining to image-based leprosy detection. The problem formulation is given in Section III. The elaborated explanation of the proposed method is given in Section IV, while the simulation results and allied conclusion are discussed in Section V and VI, correspondingly. The references used in this work are given at the end of the document.

Related Work

In the past, the authors have mainly performed invasive methods including chemical assay and histopathological analysis [20–25]. The resource dependency and time-exhaustion limit their scalability to cope with up-surging demands [26]. Though, the use of vision-based medical imaging systems has broadened the horizon for leprosy detection and classification [17–19, 26]. The diversity of spatio-textural features across leprosy lesion, ambiguity with the other skin diseases like melanoma cancers etc. make vision-based approaches challenging to guarantee reliability [16, 17]. Vision-based technologies have vital scope towards automated leprosy detection and classification tasks; provided the method ensures optimal spatio textural feature learning and classification. In recent years, numerous efforts have been made by applying improved feature extraction, feature learning and classification approaches. In this reference, to exploit deep features, Baweja et al. [27] exploited CNN feature extraction followed by entropy driven feature learning to perform Leprosy lesion classification. Authors applied Google inception-v3 concept to perform deep spatio-textural features learning to improve performance; yet the accuracy of 90% broadens the horizon for better feature and allied learning efficacy. A similar effort was made by Jain et al. [28] who applied classical deep learning method towards leprosy detection. Authors [28] could achieve the highest prediction accuracy of 91%. Despite the moderate performance, these efforts [27, 28] revealed that deep learning can yield better spatio-textural with deep intrinsic features to perform pattern learning-based prediction. Recalling that the spatio-textural features of the different skin diseases are different, Mondal et al. [29] developed a lesion-specific learning approach to classify lesions such as leprosy, Tinea Versicolor, and Vitiligo. To achieve it, authors [29] developed a customized Generative Adversarial Network (GAN) that generated synthetic data. The input features were learnt using DenseNet-121 that helped achieve the highest accuracy of 94.25%. Though, authors [26] claimed to have almost 11% higher accuracy than the features without augmentation. Consequently, it raises question for its robustness over the smaller input (unannotated) dataset. Noticeably, aforesaid approaches [26–29] applied deep learning with low kernel or dimensions, and hence to exploit more deep features, Lo et al. [30] applied deep CNN (DCNN) with pretrained AlexNet to classify bacillus image blocks. In this manner, the transferred DCNN model estimated the likelihood that each image block contained a bacillus. With a predefined probability threshold of 0.5, this approach exhibited an accuracy of 95.3%, sensitivity of 93.5%, and specificity of 96.3%. However, this method without color information could achieve the accuracy, sensitivity and specificity of 73.8%, 70.7% and 75.4%, respectively. The use of DCNN with AlexNet enabled distinguishing bacilli from other tissues with promising accuracy, signifying the superiority of the deep features [27–29]. In addition to the deep learning-based approaches, a few efforts have been made on exploiting different features including local binary patterns (LBP), Gray Level Co-Occurrence Metrics (GLCM)

[16] etc. Das et al. [16] performed multi-class classification to classify skin disease lesions as leprosy, tinea versicolor and vitiligo. To explore efficacy of the different features, authors [16] assessed features like LBP, GLCM, Discrete Cosine Transform (DCT) and Discrete Fourier Transform (DFT) towards skin lesion classification. To perform feature learning and classification, authors applied support vector machine (SVM) algorithm. Though, authors found that LBP-feature can give superior performance (with accuracy of 89.65%), yet GLCM possesses more significant spatio-textural features towards leprosy lesion classification. Vivekanadan et al. [26] too found that the use of information rich features can be better learnt over machine learning to yield reliable diagnosis or image-based Leprosy detection [31]. Authors [32] applied machine learning method to examine changes over myo-electric signals in skin lesions to diagnose disease.These methods are highly dependent on manual feature extraction, labelling and training. Pal et al. [33] developed a rotation invariant weber local descriptor for skin disease detection. They applied multiple local binary patterns feature variants towards skin lesion classification. Yet, the data used in their work could not address the level of heterogeneity and complex structure that makes it inferior towards leprosy detection and classification. Though, a few other approaches like magnetic resonance imaging have been applied to assess ulnar nerve abscess in leprosy [34]; yet its scalability to meet at hand up-surging cases seems limited. Even though numerous efforts [34–41] have been made towards human skin lesion segmentation and classification; yet majority of the existing approaches are confined to address the level of structural complexity, non-linear lesion patterns and minimal annotations. It makes such approaches confined to fulfil leprosy detection demands, especially using vision-based techniques [42, 43]. It motivates academia-industries to explore more efficient, reliable and automated leprosy detection and classification [7, 43–45].

The literatures revealed that applying diversity of spatio-textural features like GLCM and deep-features with sufficiently large (deep-dimensional) intrinsic features can enable accurate leprosy detection and classification. Though, improving computational ecosystems with superior region of interest (ROI) segmentation, ROI-sensitive feature extraction and improved classification is inevitable. in this paper a robust solution is proposed by applying evolutionary computing driven ROI segmentation followed by hybrid feature extraction and improved ensemble classification to achieve optimal leprosy detection performance. Being a first of its kind solution, the proposed model addresses every aspect of leprosy detection including non-linear lesion structure, deep high-dimensional features, local GLCM features followed by ensemble learning that enables it to deliver the best possible performance towards automated vision-based leprosy detection and allied diagnosis.

Problem Formulation

The presence of leprosy lesions on the different non-linear surfaces and patch's diversity makes annotations difficult. Under aforesaid challenges, the only measure left is to learn over the ROI-specific features. In other words, learning ROI-specific features can provide more efficient feature environment to make accurate pattern learning and classification. Majority of the existing methods including classical wavelet analysis or deep learning concepts, considering lesion non-linearity issues, authors have directly used complete lesion images for feature extraction, where the presence of non-ROI pixel regions not only imposes computational overheads but also forces machine learning classifier to learn over non-ROI feature sets that eventually causes false-positive performance. To alleviate such false positive probability, training a model over ROI-specific features is inevitable. Though, feature sets like GLCM and deep features have achieved satisfactory performance towards leprosy detection and classification, the amalgamation of these duo can yield superior results than any of the standalone methods. In sync with these scopes or allied motives, the focus is made on developing a robust leprosy detection system where the focus is made on both feature engineering as well as classification performance. To retain higher feature diversity towards learning, in this paper a large set of leprosy images representing patches on the different body parts were obtained. To include lesion diversity, it applies a significantly large image sets with different skin colors, lesion segments and its dimensions. Once obtaining the input images, to ensure ROI-specific feature extraction, heuristic driven automatic ROI segmentation method is applied. Here, the key motive is to retain only ROI specific regions while dropping the non-ROI regions to improve feature sanity. On the other hand, realizing non-linear morphological traits of the skin lesions, clustering-based segmentation is considered. Thus, to ensure automated and precise ROI-segmentation, FFCM clustering is applied that segmented ROI in each skin lesion image. Once segmenting the ROI-regions, the segmented regions are multiplied with the binary images that gives rise to the color space presentation of the ROI regions. Subsequently, to perform ROI-specific feature learning two different feature extraction methods including spatio-textural GLCM method and deep learning are applied independently. Here the key motive is to exploit both high-dimensional deep features and GLCM features altogether to ensure superior learning and hence classification. In this reference, diverse GLCM features

like energy, entropy, contrast, homogeneity, and correlation was employed. On the other hand, to ensure high dimension deep features, the proposed model exploited AlexNet-CNN features with five convolution layer and three fully connected layer. In this manner, once obtaining the high dimensional deep features and spatio-textural GLCM features, the proposed model performs horizontal concatenation to result a hybrid feature for further learning. Subsequently, the fused feature vector is applied to perform two-class classification using an ensemble learning method named random forest that classifies each input test image as the normal or the leprosy lesion. To assess depth performance, statistical approaches were taken into consideration.

System Model

The overall proposed vision-based leprosy detection model encompasses the following steps:

1. **Step-1** Data Preparation,
2. **Step-2** Heuristic FFCM driven ROI segmentation,
3. **Step-3** Hybrid Spatio-Textural and Deep Feature Extraction,
4. **Step-4** Random Forest Ensemble Learning based Leprosy Classification.

The phased implementation of the proposed method is discussed in the subsequent sections.

Data Acquisition and Pre-processing

In the past a few researchers have employed leprosy images for machine learning based diagnosis; however, no standard benchmark data are available. Moreover, existing literatures don't provide data insights, especially in reference to the patch's dimensions, skin position etc. It creates ambiguity in understanding input data characteristics to generalize the performance. In this work, a set of 220 images were collected from the different primary as well as secondary sources including hospital visits, online leprosy classification datasets etc. However, due to differences in resolution and diversity of (lesion) morphology and allied spatio-textural features, the skin-lesion images were assessed for its suitability.

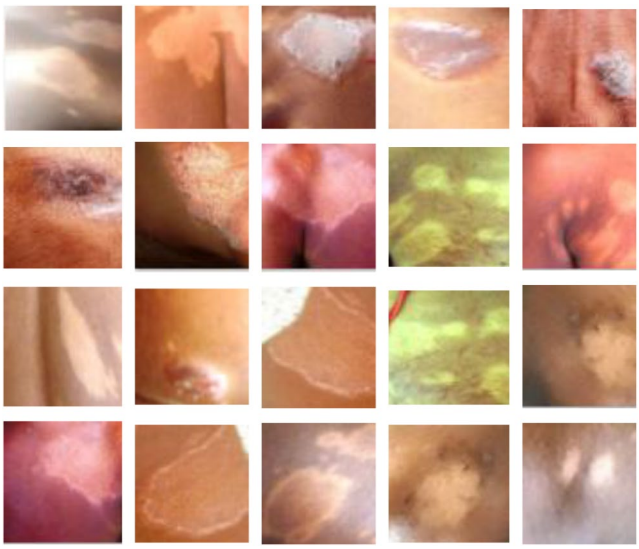


Table 1 Samples of the leprosy lesion images

Considering minimal annotations of the available data and scalability as future goal, in this work the focus was made on introducing diversity of features by selecting skin lesions from the different locations such as thigh, fingers, nose, face, limbs, palms, back side images etc. Noticeably, the data were selected from the patients of different skin colors, skin and lesion gradients etc. Here, the key motive was to use sufficiently large number of input lesions and allied features to train the model for higher accuracy. The input images were later processed for annotations, thus giving rise to a sufficiently large 450 images representing the leprosy image samples. Being a two-class classification problem, we also considered normal skin images pertaining to the different body parts. In this manner, once obtaining the set of input leprosy and normal image sets, a standard pre-processing step was followed by performing adaptive histogram equalization and image resizing. We resized overall input images in the dimension of 150×70. A snippet of the input images considered is given in Table 1.

Heuristic FFCM Driven ROI Segmentation

Once obtaining the input images and processing for the standard pre-processing steps, the proposed model executed Firefly driven Fuzzy C-Means (FFCM) clustering method to perform automatic ROI segmentation. Unlike existing methods, in this work, the emphasis is made on retaining only ROI-specific features to train the model, as inclusion of non-ROI features or allied pixel information can impact overall learning and allied classification results. Moreover, being a non-linear morphology problem, segmenting leprosy lesions can be difficult and inaccurate using classical threshold-based approaches like Otsu methods or even region growing. To alleviate this problem, in this work ROI segmentation was considered as a clustering problem where we applied FCM algorithm to segment non-linear ROI patches or lesion patches. Clustering can be a viable solution; Where it can not only make overall ROI segmentation accurate but can also alleviate the problem of manual seed-point estimation. Despite aforesaid robustness, being cluster-based approach, FCM can undergo inaccurate ROI

segmentation, especially over non-linear morphology and narrow ROI- non-ROI gap (say, differences). To alleviate this problem improving centroid estimation and update during segmentation is must. To achieve it, the proposed method applies a robust heuristic concept or algorithm named Firefly algorithm that in sync with FCM performs ROI segmentation.

ROI Detection

Being a cluster driven approach, to improve segmentation efficiency over non-linear morphology, the proposed method intends to cluster different pixels over the skin lesions possessing similar spatio-textural cues, by minimizing intra-cluster distance. Noticeably, the proposed model not only alleviates any need of supplementary manual seed-point definition, but also suppresses post-segmentation tasks such as blobs removal or connected component analysis (CCA). In this paper, FCM algorithm was applied onto each input images to perform clustering, where Firefly heuristic helped FCM to improve cluster centroid(s) and hence ROI-segmentation. FCM-based Lepro-specific ROI segmentation is discussed subsequently.

Fuzzy C-Means

Let, x_i be the pixel elements in an input image. Then, the proposed FCM clustering method is expected to group the set of n patterns or instances spread over the input image and given as $x = \{x_1, x_2, x_3, \ldots, x_n\}$. Here, $x_{i\in d}$ represents the feature vector possessing d real-valued estimations representing features x_i of the input pattern. Unlike other clustering approaches like K-Means methods, Fuzzy in FCM helps defining membership functions (MF) that eventually enhances clustering over non-linear patterns (say, $x_{i\in d}$). Moreover, the efficacy of FCM to enable participating instances belonging to the multiple clusters and label them accordingly strengthens it to yield higher accuracy. Amongst the two variants of FCM, the Hard Fuzzy model splits input data x into the different groups G_1, G_2, $G_3 \ldots$, G_c, where one pattern belongs to only one cluster. Such patterns of instances are categorized and labelled into more than two non-overlapping clusters. On the contrary, in case of Soft Fuzzy method an instance x can also be the part of these clusters. For leprosy lesion or ROI detection problem, there can be the multiple patches or morphologically ambiguous elements available in both ROI as well as non-ROI feature space. To alleviate this problem, soft-Fuzzy based clustering can be vital. Thus, soft-Fuzzy clustering model, our proposed method estimates the output of the clustering in the form of a membership matrix, often called Fuzzy Partition Matrix $U = [u_{ij}]_{(c,n)}$. In the proposed model, we obtained Fuzzy membership function (FMF) for all groups using Eq. (1).

$$M_{FCN} = \left\{ u \in R^{c,n} \left| \sum_{j=1}^{c} U_{ij}, 0 < \sum_{j=1}^{n} U_{tj<n} \right. \right\} \tag{1}$$

In (1), $u_{ij} \in [0,1]$ states the FMF for the $i-th$ pattern in the $j-$ th cluster. Moreover, it retains the condition $U_{ij} \in [0,1]$, provided $1 \leq j \leq c$ and $1 \leq i \leq n$. Functionally, in this work, FCM is executed over the complete input instances that intends to reduce an objective function iteratively so as to achieve optimal cluster (2).

$$J_m = \sum_{j=1}^{c} \sum_{i=1}^{n} u_{ij}^{m} \parallel x_i - v_j \parallel \tag{2}$$

In (2), $\{v_j\}_{j=1}^{c\|.\|}$ represents the cluster's centroid. Noticeably, in FCM, the cluster's centroid is derived as the mathematical inner-product norm from x_i to the $\mathrm{j}-\mathrm{th}$ cluster centers. Here, the proposed model applies FMF characterizing the Fuzziness level of each pattern by applying the weighting exponent $m \in [1, \infty]$. In the proposed FCM clustering method, the following steps were employed.

- Start clustering of input pixel instances by employing c randomly selected centers.
- Calculate the membership value for each pixel in each formed cluster.
- Re-estimate the centroid value in each iteration by reducing an objective function.
- Repeatedly process membership metrics estimation (3) and centroid update and continue it till there is no change in centroids.

In this work, to calculate fuzzy membership of each ROI-specific pixel (in each group) Eq. (3) was applied.

$$u_{ij} = \frac{1}{\sum_{k=1}^{c} \left(\frac{\|x_i - v_j\|}{\|x_i - v_k\|} \right)^{\frac{2}{m-1}}} \tag{3}$$

In native FCM based clustering methods, the centroid of each group is estimated and updated by applying Eq. (4).

$$v_j = \frac{\sum_{i=1}^{n} u_{ij}^{m} . x_i}{\sum_{i=1}^{n} u_{ij}^{m}} \tag{4}$$

Thus, applying above stated method FCM applies centroid information (4) to perform clustering; yet its efficacy can be limited over non-linear morphological diverse and complex lesions [15]. To cope up with non-linear pattern segmentation with high ambiguity likelihood, the centroid update problem can be solved as a convex optimization problem,

for which a suitable heuristic approach can be considered. In this reference, the proposed work applied Firefly method, especially to update centroid information efficiently so as to accomplish ROI-specific segmentation. A snippet of the proposed Firefly algorithm for FCM centroid update is given in the subsequent section.

Firefly Algorithm Based FCM Centroid Optimization

Despite the fact that FCM clustering performs superior over the classical K-Means clustering, because of its adaptive clustering. This makes FCM suitable clustering method for image analysis and segmentation under unclear lesion morphologies. It enables FCM to be used in numerous purposes including target detection and classification, image analysis, text classification etc. However, as stated above, applying static or inferior cluster centroid might force it to yield inaccurate or skewed clustering. To alleviate this problem, in this paper Firefly algorithm is applied to optimize centroids in each cluster. Unlike other classical heuristics which require large number of iterations and hence computations, the ability to yield sub-optimal solution with fewer computation makes Firefly algorithm suitable towards our proposed clustering optimization or allied centroid optimization problem. Firefly algorithm minimizes the inter-cluster distance to enhance centroid information that helps improving eventual clustering accuracy. Functionally, Firefly algorithm is developed on the basis of the Firefly's light flashing and light-sensitive movement behavior. This method applies light-flash behave as a signal to attract nearby fireflies. In this manner, hypothesizing each pixel elements as Firefly, the proposed model tries to attract other Fireflies (i.e., pixel instances) to the same cluster iteratively that improves overall clustering accuracy. Before discussing the proposed FFCM method for leprosy ROI segmentation, a snippet of the rule methods used is given as follows:

Rule-1 Firefly being unisexual in nature gets attracted towards another firefly. Noticeably, this behavior is independent of any gender.

Rule-2 Fireflies attract other Firefly in sync with their corresponding illumination intensity, where they attract other in reverse proportion to their search spaces. In this phenomenon, a Firefly flashing brighter often attracts other possessing lower brightness. On the other hand, the one with higher inter-instance distance influences lower than the others. In other words, it follows the condition, where signifies the square of the distance. Thus, higher would cause lower attractiveness. If, a Firefly with no neighboring Firefly with higher brightness would move arbitrarily towards the one with higher intensity.

Rule-3 A firefly possessing the brightest intensity cannot be attracted and therefore remains moving arbitrarily.

Firefly algorithm was applied to estimate the optimal set of centroids in each image to segment the ROI regions. Here, it intends to estimate the centroid in a $N-$ dimensional search space by using correlation in between the cost function and the inter-pixel distance. Initially, FFCM distributes population randomly across the search space. It comprises two key processes, first the calculation of the intensity differences (here, we define it as the objective function). In sync with the objective function, the firefly with lower or higher brightness attracts other Firefly having lower or higher brightness. This approach helps aggregating the pixel elements having similar pattern to a centroid and thus forms cluster(s). Let, there are $_n$ fireflies constituting search-space and $_{x_i}$ states the solution for $i-th$ firefly. In this manner, $f(x_i)$ be the fitness value for $_x$, while allied brightness be I. The aforesaid function measures the actual position of i. We measure the intensity of the Firefly by using Eq. (5).

$$I_i = f(x_i)\, 1 \leq i \leq n \tag{5}$$

Once estimating the intensity of Firefly, this algorithm executes swarm movement where Fireflies attack other fireflies, where the attractiveness of a firefly is proportional to the intensity of the neighboring fireflies. To be noted, every Firefly possesses certain definite attractiveness β that solely relies on the distance values r_{ij} existing between the instance i and j across the swarm.

$$r_{ij} = \| x_i - x_j \| \tag{6}$$

$$\beta\ (r) = \beta_0 exp^{\{-\gamma\, d(i,j)^2\}} \tag{7}$$

In (7), β_0 represents the attractiveness factor where both fireflies are at the same position (i.e., $r = 0$). $_\gamma$ states the light absorption-coefficient, assuming that the Firefly j possesses higher brightness than the other Firefly i. Thus, i traverses toward j and the distance traversed is calculated as per (8). In (8), $rand$ refers an arbitrary number generator generating random numbers in the range of 0 and 1.

$$x_i(t+1) = x_i(x_j - x_i) + \alpha\ (rand - 0.5) \tag{8}$$

In this manner, the FA algorithm estimates the optimal position of the Firefly, signifying the centroid of the ROI block. Equation (9) presents the set of possible array A, while $_{a_i}$ be the centroid where $a_i \in A$. In (9), $_{s_i}$ represents the

centroid of each cluster in reference to the feature elements $d\{a_1, a_2, \ldots, a_d\}$. These centroid values are updated to the FCM algorithm to perform clustering where it keeps updating the centroid till there is no change in centroid value and hence sub-optimal solution.

$$A = (s_1\{a_1, a_2, \ldots, a_d\}, \; s_2\{a_1, a_2, \ldots, a_d\}, \; s_3\{a_1, a_2, \ldots, a_d\}) \tag{9}$$

To implement Firefly, we assigned number of populations as 25 ($_n = 25$), while the number of iterations (say, stopping criteria) was fixed at 100. Meanwhile, we assigned $\beta = 1$ and $_g = 1$. We applied Rosenbrock function as objective function.

In FFCM, the fitness value of Firefly algorithm signifies how good or bad, an image pixel is to become the centroid. In other words, here, fitness value refers the probability of a pixel element to become the centroid of each cluster. In this method, a firefly possessing higher intensity is hypothesized to have low fitness value that enables attracting other fireflies and controls the movement. Thus, when examining two different fireflies $_a$ and b, when b possesses the intensity more in comparison to the value of $_a$. In sync with this scenario, $_a$ traverses towards b, and this approach continues over a definite number of iterations. Thus, it generates updated population (say, sub-optimal solutions) $a^{'} = a_1^{'}, a_2^{'}, a_{31}^{'}, \ldots, a_N^{'}$, by using $f(a^{'})$. This work applied Rosenbrock method as the cost function to achieve fast convergence. It applies Rosenbrock valley function. Noticeably, identifying valley region is a complex task as not all solutions can have unit ('1') value (10).

$$f(x, y) = (a - x)^2 + b\left(y - x^2\right)^2 \tag{10}$$

To achieve global minima at $(x, y) = \left(a, a^2\right)$, Rosenbrock function requires fulfilling the condition $f(x, y) = 0$. Functionally, it is defined in such way that it results $a = 1$ and $b = 100$. In this manner, with $a = 0$, it results a symmetric function and minimum at the origin. In this work, a multi-dimensional generalization approach was applied to accomplish global minima by applying Rosenbrock function (11).

$$f(x) = \sum_{i=1}^{N-1} 100\left(x_{i+1} - x_i^2\right)^2 (1 - x_i)^2 \tag{11}$$

In (11), $X = [x_1, x_2, \ldots, x_N] \in \mathbb{R}^N$ encompasses unit minima with $N = 3$ $(at\,(1,1,1))$, while it contains two minima conditioned at $4 \leq N \leq 7$. It achieves expected global minima of all ones, especially with the local minima near $(x_1, x_2, \ldots, x_N) = (-1,1, \ldots, 1))$. In this work, we achieved this condition by assigning the gradient of the function as zero, provided the function (11) remains rational function of $_x$. Thus, applying Rosenbrock function as objective function FFCM method estimated optimal

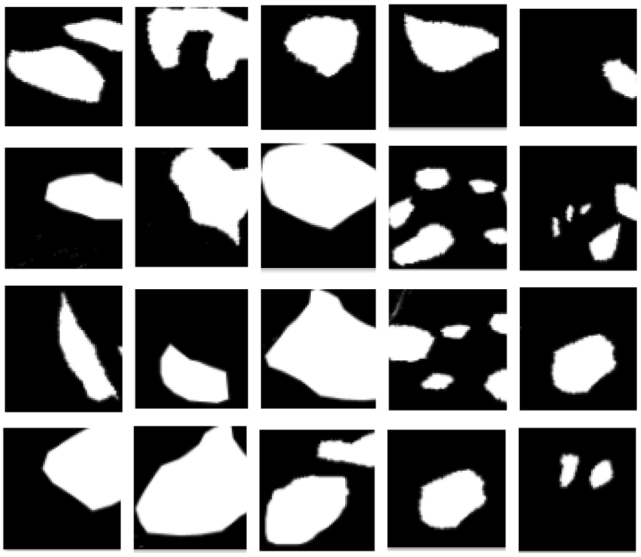


Table 2 ROI segmented regions (binary)

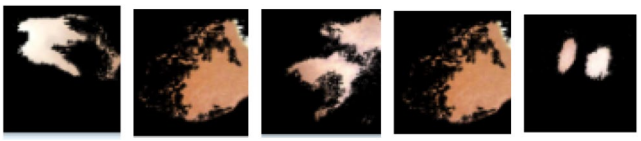


Table 3 Segmented ROI in colour space

set of centroids and corresponding clusters representing the lesion-specific ROI regions. In this work, FFCM was applied over each input image where it segments ROI. Subsequently to extract features in RGB frame, we multiplied segmented binary image (i.e., lesion ROI) with corresponding RGB channel pixels and thus ROI-region was obtained in RGB space for each input image (Table 2-3).

Once estimating ROI-specific lesion outputs in RGB frame, we performed spatial textural feature extraction. The details of the feature extraction method applied in this work is given in the subsequent section.

Hybrid Spatio-Textural and Deep Feature Extraction

Unlike classical approaches, in this paper both local spatio-textural features as well as deep features have been exploited to perform leprosy prediction. To achieve it, GLCM and AlexNet deep methods were applied to extract two distinct features from the segmented leprosy lesions. To be noted, to retain optimal spatio-textural feature traits, once segmenting the ROI, we performed pixelwise multiplication. In other words, the proposed model multiplied RGB channel information (say, RGB pixel values) over the segmented regions. In this manner, the proposed model obtained RGB spatio-textural feature space pertaining to the precisely segmented ROI region. This as a result helped retaining only those

local features possessing ROI-specific spatio-textural cues. Performing pixel-wise multiplication over the segmented ROI, we executed hybrid spatio-textural and deep feature extraction, using GLCM and AlexNet-CNN, respectively.

GLCM Based Statistical Feature Extraction

In this work, spatio-textural volume analysis approach was considered to extract different spatio-textural feature extraction. In this method, the motive was to collect features from the ROI-specific regions in 3D feature map (say, 3D-RGB data block). More specifically, distinct spatio-textural features were extracted from each lesion patch and allied ROI spaces. Here, the use of spatio-textural textural can provide local motion pattern which can be vital towards leprosy identification. In sync with these motives, the proposed model exploited the different features including statistical and signal domain textural features (SDTF). To achieve it, GLCM was employed. For statistical feature, the gray scale values present between the ROI pixels and neighboring pixel. In this paper, the high-degree statistical features including the 1st, 2nd -dimensionalstatistical features like contrast, variance etc. were obtained. Subsequently, other textural features were obtained. The extracted features were amalgamated together to give rise to a console feature set called GLCM. We hypothesize that the inclusion of signal domain textural features such as energy distribution can be applied to perform texture-classification, which can help leprosy detection and classification mechanisms. Considering very fine and complex micro-level feature information (or cue), we hypothesized that the aforesaid information can be present in highly fine-grained granular form distributed across the ROI regions. Moreover, since we performed feature extraction over merely the ROI specific regions or lesion areas, the extracted feature was defined as a matrix representing the matrix of pixel intensities $I(x, y)$. The calculation of the likelihood matrix $P_{i,j}$ for each input image, signifying the intensity difference between the pair of image pixel i and j was expected to be employed so as to detect motion pattern(s). In the proposed model, the gray-scale represents the pair relation towards one direction. The gray-scale values are then transformed to yield transpose matrix followed by the symmetric matrix S by performing addition of the gray-scale information with corresponding transpose that finally define the cumulative relationships in a direction. In this work (12) was applied to normalize the relationship matrix S that subsequently obtained the probability matrix P.

$$P_{i,j} = \frac{S_{i,j}}{\sum_{i,j=0}^{N-1} S_{i,j}} \tag{12}$$

Now, once estimating the matrix $\mathrm{P_{i,j}}$ (12), different STTF features, often called low-level textural pattern or textural signature were obtained. In this work, we obtained five different GLCM features including contrast, homogeneity, entropy, energy and correlation as textural signature to perform final spatio-textural feature estimation. A snippet of the different spatio-textural features is given in the subsequent section.

Contrast Typically, contrast states the variation of the gray scale value over an input image in the form of contrast, homogeneity, dissimilarity etc. The pixel pairs available in diagonal of $P_{i,j}$(12) represent higher difference in contrast or the corresponding gray scale values. Here, contrast states the total variations of the pixel intensities across input lesion. Functionally, the disturbances present in the spatio-textural texture can be identified by performing statistical measurement and corresponding (textural) uniformity assessment.

Homogeneity In GLCM features, homogeneity refers the Inverse Different Moment (IDM), signifying low contrast. The proposed method used the Eq. (13) to measure homogeneity in each input image.

$$HOM = \sum_{i,j=0}^{N-1} \frac{P_{i,j}}{1 + (i-j)^2} \tag{13}$$

Noticeably, the typical GLCM features encompass the values uniformly distributed over the grids. In case GLCM possesses merely a small fraction of the large magnitude, the contrast is stated to be high and therefore low homogenous. On the contrary, with the uniform magnitude across the image, it refers low contrast and hence high homogeneity. In this manner, contrast is estimated over an image using (14).

$$CONT = \sum_{i,j=0}^{N-1} P_{i,j}(i-j)^2 \tag{14}$$

Energy To estimate energy distribution over the target lesion region or the input sample, angular second moment information (ASM) was applied. ASM assesses the rotational acceleration throughout the input legion or ROI space, which can be derived as per (15). In general, the values of ASN increases over the uniform gray-scale) value distribution.

$$ASM = \sum_{i,j=0}^{N-1} {P_{i,j}}^2 \tag{15}$$

Thus, once estimating the value of (15), we applied the model derived in (16) to measure energy feature.

$$ENR = \sqrt{ASM_{i,j}} \tag{16}$$

Entropy Unlike energy (16), entropy feature states the extent to which the gray values are distributed non-linearly over input image or allied ROI. Typically, entropy increases as per rise in non-linearity or irregularity in pixel distribution. Mathematically, we applied (17) to estimate entropy information.

$$ENT = \sum {}_{i,j=0}^{N-1} P_{i,j}\left(-lnP_{i,j}\right) \tag{17}$$

Correlation In addition to the above discussed features, this work also exploited the descriptive statistics. More specifically, in this work, three descriptive statistics including Mean, Variance and Correlation were obtained. These features represent the statistic of pixel pair gray scale relation and therefore it doesn't mean merely the classical gray scale values. In this manner, the pixel-pair relationship enables aforesaid features suitable to detect target class for further classification or annotation. The Eq. (18) was applied to measure two GLCM Mean values. To be noted, because of the symmetric characteristics of the likelihood matrix (12), these two mean values can turn out to be equivalent.

$$\mu_{\,i} = \sum {}_{i,j}^{N-1} i\left(P_{i,j}\right)\mu_{\,j} = \sum {}_{i,j}^{N-1} j\left(P_{i,j}\right) \tag{18}$$

The Eqs. (19) and (20) are applied to measure the Variance and the Standard Deviation, correspondingly.

$$\sigma_{\,i}^{\,2} = \sum {}_{i,j}^{N-1} P_{i,j}(i-\mu_{\,i})^2 \sigma_{\,j}^{\,2} = \sum {}_{i,j}^{N-1} P_{i,j}\left(j-\mu_{\,j}\right)^2 \tag{19}$$

$$\sigma_{\,i} = \sqrt{\sigma_{\,i}^{\,2}}\sigma_{\,j} = \sqrt{\sigma_{\,j}^{\,2}} \tag{20}$$

Once estimating the mean and variance information, correlation feature was obtained as per (21).

$$CORR = \sum {}_{i,j}^{N-1} P_{i,j}\left[\frac{\left(i-\mu_{\,i}\right)\left(j-\mu_{\,j}\right)}{\sqrt{\left(\sigma_{\,i}^{\,2}\right)\left(\sigma_{\,j}^{\,2}\right)}}\right] \tag{21}$$

Now, once extracting above stated GLCM features (i.e., Contrast, Energy, Entropy, Homogenity, Correlation, Variance, Mean and Standard Deviation), we performed horizontal concatination that yielded a cumulative spatio-textural features. The combined GLCM features used are defined as per (22).

$$GLCM_{Feat} = [CONT, ENE,\ ENT, HOMO, CORR,\ VAR,\ Mean,\ SD] \tag{22}$$

Thus, we extracted spatio-textural GLCM features (22) for each input sample to be further trained and learnt to perform leprosy lesion classification.

Unlike classical spatio-textural feature driven leprosy detection and classification systems, in this work a hybrid fetaure driven leprosy classification model is developed. Here, we hypothesized that unlike classical standalone feature driven approaches, the strategic amalgamation of both GLCM and deep features can be of great significance. Though, a few existing approaches applied deep learning methods for leprosy lesion classification; however, lacked high dimensional deep features, which could have enabled more reliable and accurate prediction. To address this problem, in this paper, AlexNet deep network was used. Here, AlexNet was designed with five convolutional layers (CONV) and two fully connected (FC) layers, where the final neurons with 4096 dimensional features were applied to perform lesion classification. We applied AlexNet based 4096 dimensional features in conjunction with the above derived features (22) to perform leprosy prediction.

AlexNet- Driven Feature Extraction

As stated above, this research focused on exploiting both spatio-textural (descriptive statistics) features along with the high-dimensional deep features. To achieve deep feature extraction, AlexNet with three CONV and two FC layers was designed. Unlike classical convolutional neural network that exploits merely 256-dimensional features, AlexNet extracts 4096-dimensional features to perform leprosy lesion classification. Here, the use of 4096-dimensional features provides sufficiently deep (intrinsic) features to perform leprosy lesion classification. In this work, we designed AlexNet model with five convolutional layers (CONV) and three fully connected layers (FC). However, realizing the high dimensional (4096 dimensional) FC6 and FC7 (Fig. 1) was applied. Since, FC8 contains merely 1024-dimensional feature vector to perform learning, we considered FC6 fully connected layer as classifier layer to perform lesion prediction. As depicted in Fig. 1,AlexNet embodies native structure with five CONV layers (i.e., CONV1, CONV2, CONV3, CONV4 and CONV5), and two FC layers (FC6 and FC7).Since, the proposed work considered merely 220 images for leprosy detection and classification and therefore it was employed over normal central processing unit (CPU)

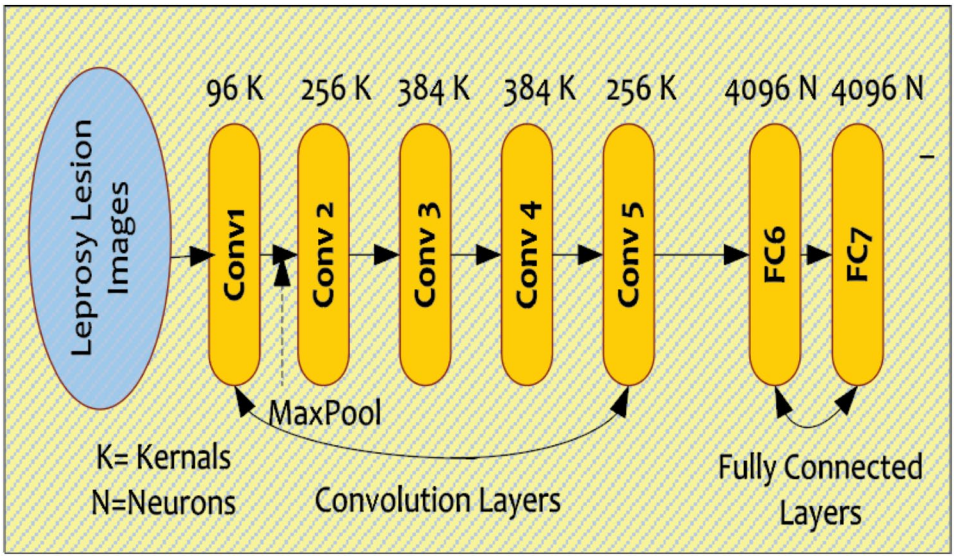


Fig. 1 AlexNet CNN model

Table 4 AlexNet deep learning model configuration

Layer	Layer Name	Specific Operation
1	Input	150×70 with zero-center normalization
2	CONV	150×70 convolutions with stride [3 3] with padding zero.
3	ReLu	ReLu regularization activation
4	Max-Pooling	3×3 Max-Pooling with stride [4 4] and padding [0 0 0 0]
5	FC	FC6
6	Drop-out	50%
7	Learning Rate	0.0001
8	Optimiser	ADAM learning
9	CONV	CONV1-CONV5
10	Epoch	200

configuration. The sequential CONV implementation with corresponding convolutional filters enables extracting deep feature with significantly vital features to perform feature learning towards leprosy detection and classification.The configuration parameters for the deployed AlexNet model are given in Table 4.

A snippet of AlexNet architecture and its implementation is given as follows:

Input Layer In the deployed transferable deep learning model, the input segmented regions (i.e., ROI specific regions) were resized to 150×70 and were fed directly to the input layer. The first layer of AlexNet executes convolutional filters (i.e., horizontal and vertical filters) to generates CONV1 features to be processed for max-pooling. Noticeably, the Max-pooling layer performs consolidattion of the extracted CONV fetaure maps and thus retains high reolution feature for further tasks. To improve performance, the proposed model executed feature scaling followed by averaging subtraction or mean subtraction in which the images are resized (here, we retained the same dimension of $150 \text{ x } 70$at each CONV layer.

CONV Layer CONV amalgamates two different filters applied distinctly as the horizontal and the vertical filters to learn patterns and extract deep features over input lesion spaces or ROI per sample. CONV retrieves as the output signifying the feature map specific to the ROI inputs (here, lesion ROI). In this process, every encompassing neuron in the extracted feature map share equivalent weights (W) and bias (b) values. The consideration of the similar weights and bias values strengthen each neuron to detect the similar features or the ROI-specific features or cues. CONV feature maps used distinct sets of weights and biases so as to obtain the local features. CONV filters the input images so as to estimate a single feature map representing the CONV layer's output. As depicted in above figure in the deployed AlexNet model a total of five CONV layers encompassing CONV1, CONV2, CONV3, CONV4 and CONV5 were applied.

Moreover, it applied the zero padding with the stride value of 3. This work retained the native structure with 96-kernels for CONV1, 256-kernels for CONV2, 384-kernels for CONV3 and CONV4-384, while CONV5 applied 256 dimensional kernels to perform feature extraction.

Max-Pooling In the deployed AlexNet network, Max-pooling is applied as the feature selection layer that intends to reduce the spatial resolution of every feature map as retrieved from each CONV layer. Max-pooling layer intends to reduce computation of the hyper-parameters and therefore enables swift computation to meet timely computational demands.

To achieve it, the proposed model executes local averaging and sub-sampling. It not only helps in improving computation but also alleviate any probability of over-fitting while learning.Max-pooling layer was used to retrieve translation-invariant feature representation. Functionally, in this work it down-sampled the latent representation that helped AlexNet model to avoid irrelevant or less significant solutions and thus avoid learning to reduce computational overhead. The deep learning model applied unit Max-pooling layer after each CONV layer, where, each layer encompassed 3×3 receptive field with stride 4.

ReLu In the deployed AlexNet deep learning model ReLu was applied as an activation function. In fact, ReLu encompasses nonlinear element-wise operator which acts as a layer in deep learning architecture. In this work, a total of three ReLu layers were applied. Here, with $_y$ as input, ReLu estimated the output for neuron $q(y)$ as $_y$ conditioned at $y > 0$. On the contrary, for $y <= 0$, it resulted output as $(\delta \times y)$, where δ be the condition signifying whether the non-positive elements are needed to be ignored by multiplying it with a slope (ex. 0.01…) or defining it as a fixe value zero. In this work, we defined $\delta = 0$that enabled ReLu to act as a native or default layer with function$q(y) = max(0, y)$.

FC Layers In deep learning model FC layer acts as a classification layer and performs high-level reasoning over the concatenated feature vector. In this work, FC layer merely concatenates the feature vector (obtained from the different neurons) from the trailing layer and eventually maps it to the connected neurons to give rise to a one-dimensional (feature) vector. In this work, we merely use the concatenated feature or the FC layer's feature vector as input to perform further two-class classification. Typically, FC layers in AlexNet possesses three FC layers named FC6, FC7 and FC8, which possesses 4096, 4096 and 1024 dimensional features, correspondingly. Considering intrinsic feature significance and high-dimensional rich feature, we considered FC6 feature vector as the final deep model's feature (vector) to perform further learning and classification. Once obtaining FC6 features, we performed concatenation of both GLCM features as well as AlexNet features that yield (23).

$$Hybrd_{Feat} = [GLCM_{Feat}, CNN_{FC6}] \tag{23}$$

The final combined feature was processed for classification using an ensemble machine learning classifier named, Random Forest algorithm.

Random Forest Classifier

Random forest being an ensemble machine learning method encompassing multiple tree-based learners and classifiers. In the deployed tree-structure, every tree retrieves distinct vote (1-Leprosy, 0-Normal), signifying the best possible class to label each sample as normal skin or leprosy. Consider, N be the total training samples, then a unit image (and allied feature) possessing N cases are arbitrarily chosen from the input data.The selected samples are later used as training sample so as to constitute a new tree. In case of M inputs, the optimal split would be $_m$, which is obtained for each node. Though, in this work, we maintained $_m$ as fixed value (here, $m = 50$) to constitute forest. Noticeably, unlike traditional decision tree methods and other machine learning approaches (Ex. Support vector machine, neural network, Naïve Bayes, regression, etc.) the Random Forest method requires lower parameters to be measured during classification that makes computation more efficient, especially over real-time tasks. Thus, the amalgamation of the different trees to form random forest can be defined as (24).

$$\{h(x, \theta_k),\ k = 1,2, \ldots i \ldots\} \tag{24}$$

In (24), h is the classifier function, while the random vector distributed uniformly is given by $\{\theta_k\}$. As stated above, in the random forest method, each encompassing tree comprises unit vote signifying the best likely class label for an input lesion image $_x$. In this work, we used a bootstrapped training (image) samples to train every tree while performing forest (say, growth stage). To improve reliability, we considered 80%of the samples for the training, while remaining 20% were used for inner cross-validation so as to validate the performance. Thus, performing above stated approach, each input leprosy lesion image was classified as normal image or the leprosy diagnosed or detected.

Results and Discussion

In this work, a novel leprosy detection solution was proposed, where the emphasis was made on both leprosy region or ROI segmentation as well as ROI-specific hybrid feature learning to improve accuracy and reliability of the system. Being a vision-based solution, skin lesion samples were collected from the different sources including primary data (i.e., data collected from the hospitals, leprosy disease dedicated hospitals etc.) as well as secondary sources such as online image websites, leprosy centric data platforms etc. Since, the leprosy lesions can be found over the different body parts with often non-linear shapes and textural differences, to ensure diversity of samples, we considered a total of 220

leprosy specific images. Noticeably, the samples considered were presenting leprosy lesions on the different body parts, such as hand, upper palm, limbs, thigh, face, etc. On the other hand, to ensure diversity of spatio-textural heterogeneity we considered samples from the different body parts as well as the persons with the different colors and diverse lesion shapes. The key purpose of retaining the lesions with different shapes, size and orientation was to assess whether the proposed FFCM segmentations model could detect the ROI accurately or not. Moreover, the intend was to retain merely lesion-specific ROI for feature extraction and learning, as training any model over irrelevant (say, non-ROI) feature space can make prediction ambiguous and prone to the false positive. In this process, the input leprosy images (as well as the normal skin lesions pertaining to the different body parts) were at first converted into Gray-level images from the RGB space, which was followed by pre-processing techniques like histogram equalization and two-dimensional median filtering. In addition, we performed Z-score normalization over the filtered input images so as to alleviate any probable over-fitting problem. To perform FFCM segmentation, FCM was executed with the number of clusters as two. This is because, we wanted to segment overall image with background as one cluster, while leprosy lesions as another cluster. Fuzzification parameter was defined as 2, while the maximum number of iterations was fixed at 100. We employed Rosenbrock function as the objective function to perform Firefly-based lesion-region segmentation. In this reference, the total number of Fireflies (say, population) was fixed as 2, while light absorption coefficient and attraction coefficient base values were defined as 1 and 2, respectively. The lower and upper bound (say, decision variables during optimization) were defined as -10 and 10, respectively. We considered mutation coefficient and related damping ratio as 0.2 and 0.98, correspondingly. The value of the uniform mutation range was fixed as unit value 1. Now, executing Firefly algorithm in conjunction with the FCM clustering, we segmented ROI specific regions which was saved and resized to 256×300 dimension. Thus, the resized images were further processed for feature extraction using GLCM and AlexNet methods distinctly. In reference to the GLCM features, we extracted key descriptive features including homogeneity, entropy, energy, mean, variance and standard deviation. Here, we horizontally concatenated the GLCM features which was later fused with AlexNet deep learning feature. As second feature extraction tool we applied AlexNet deep learning model with 5-CONV and two FC layers. Here, we obtained 4096-dimensional feature at FC7. Here, we applied ADAM learning methods with 200 number of epochs and 100 iterations so as to achieve AlexNet deep features. Thus, the extracted GLCM and FC7 features were fused. The final fused features were later used to perform ensemble learning-based two-class classification. Thus, the Random Forest algorithm classified each skin lesion as the normal image or leprosy lesion. To assess performance in this work, statistical performance parameters including leprosy prediction accuracy, precision, recall and F-score. To achieve these parameters, we obtained confusion metrics in the form of true positive (TP), true negative (TN), false positive (FP) and false negative (FN). A snippet of the statistical parameters used for performance characterization is given in Table 5.

Table 5 Performance parameters

Parameter	Mathematical Expression
Accuracy	$\frac{(TN+TP)}{(TN+FN+FP+TP)}$
Precision	$\frac{TP}{(TP+FP)}$
Recall	$\frac{TP}{(TP+FN)}$
F-Score	$2.\frac{Recall.Precision}{Recall+Precision}$

We assessed the leprosy-detection and classification performance for both GLCM as well as hybrid features. To be noted, a recent study hypothesized that GLCM features can yield leprosy prediction accuracy of almost 88%, with classical cumulative image-feature learning. In other words, the existing state-of-art method hypothesized that extracting GLCM features from a complete skin lesion can enable a CAD to exhibit 88% of the leprosy prediction accuracy. However, the existing method considered the features from across the image (not only the ROI part). Therefore, in this case, the impact of non-ROI features on learning outputs can't be ignored. The similar approach was followed by Baweja et al. [27], who employed complete image possessing leprosy lesions to perform CNN-based feature extraction. The highest accuracy observed by Baweja et al. [27] was 93%. On contrary to these state-of-arts, in this paper we at first obtained ROI-specific hybrid features encompassing both GLCM as well as AlexNet deep features, which can be hypothesized to carry more information rich feature vector for learning and classification.

Intra-Model Assessment

In this approach, we assessed the performance with GLCM features and the hybrid features (i.e., GLCM and AlexNet). Here, our key motive was to assess whether the inclusion of hybrid feature yields superior performance over merely GLCM driven prediction model. The experimental outputs in the form of accuracy, precision, recall and F-score are given in Table 6.

The above results (Table V) present the statistical performance by GLCM features and the proposed hybrid feature (i.e., GLCM and AlexNet features altogether). to achieve this relative performance (Table V), we applied common

Table 6 Intra-model performance assessment (average)

Features/Model	Accuracy (%)	Precision (%)	Recall (%)	F-Score
GLCM	92.7	89.5	92.1	0.9402
Hybrid Feature (GLCM + AlexNet)	96.6	99.7	95.8	0.9771

pre-processing algorithms (i.e., histogram equalization, median filtering and Z-score normalization) over the input sample(s). Moreover, the classifier employed was the Random Forest algorithm, applied as common classifier with both features. The simulation results show that GLCM as standalone feature can achieve average (leprosy detection and classification) accuracy of 92.7%, precision of 89.5%, recall of 92.1% and F-score of 0.9402. On the other hand, our proposed hybrid feature-based leprosy prediction model showed better efficacy with accuracy 96.6%, precision 99.7%, recall of 95.8% and F-score of 0.9771. The relative performance analysis exhibits that the proposed hybrid feature (i.e., spatio-textural descriptive GLCM statistics along with the high-dimensional AlexNet features) delivers superior performance than the GLCM as standalone feature for leprosy prediction. This can be mainly because of the fact that GLCM at one had provided vital local descriptive statistics of the spatio- textural features, while AlexNet with 4096 dimensional (deep) features helped providing the optimal intrinsic features to guarantee optimal feature learning and classification. Thus, the results obtained confirm that the use of hybrid spatio-textural deep features can yield superior and more reliable performance towards vision-based touchless leprosy detection and classification task. In this manner, the research questions (RQ2 and RQ3) as framed in the Section III are found affirmative stating that the use of hybrid spatio-textural and high dimensional deep features be effective towards leprosy detection and classification (RQ2). Since, the proposed model was executed for two-class classification (i.e., normal skin image and leprosy detected) using Random Forest algorithm and hence affirmative results are acknowledged towards RQ3, stating that the hybrid spatio-textural and deep features with the Random Forest classification can yield superior performance towards leprosy detection and classification. Now, considering the average performance of the proposed Hybrid Spatio-Textural Deep Feature driven leprosy detection and classification model, we have compared its efficacy over the other state-of-art vision-based leprosy detection and classification techniques.

Inter-Model Assessment

Very few efforts have been made towards vision-based leprosy detection and classification systems. However, exploring in depth, we identified a few state-of-art approaches applying local binary patterns [16], GLCM [16], and convolutional neural network (CNN) for leprosy image classification [27]. To be noted, the depth assessment reveal that the almost all existing approaches directly feed leprosy lesion images containing both ROI as well as non-ROI space to perform feature extraction and learning. In these approaches, the likelihood of biased learning (as the presence of ROI space can be quite smaller in comparison to the non-ROI spaces across 2D image(s)) can't be ignored and hence such approaches might give false positive performance. Unlike these approaches, in this paper FFCM assisted ROI-segmentation was performed where only ROI-specific regions were retained for further feature extraction. This as a result helped alleviating or reducing irrelevant training and hence can be considered to have impact on better performance. On the other hand, a recent work by Baweja et al. [27] who applied CNN for leprosy detection and classification considered whole input image as feature space to train the model. Moreover, it could extract merely 256-dimensional features. In comparison to it, our proposed AlexNet model extracted 4096-dimensional features that can provide superior or information rich feature vector for learning and hence eventual leprosy detection and classification. Thus, these qualitative assessments reveal superiority of the proposed leprosy detection and classification system. To examine relative performance, we compared performance of the proposed model with the other state-of-art vision-based leprosy detection. The relative performance of the different approaches is given in Table 7.

Table 7 Inter-model performance assessment

References	Features	Accuracy (%)	Precision (%)	Recall (%)	F-Score
Lo et al. [30]	DCNN	73.8	-	93.5	-
Mondal et al. [29]	GAN + DenseNet-121	94.25	94.0	90.4	0.922
	MobileNet V2	81.6	-	-	-
	ResNet-101	87.35	-	-	-
	GoogLeNet	89.65	-	-	-
	DenseNet-169	93.67	-	-	-
	DenseNet-121	94.25	-	-	-
Das et al. [16]	LBP	89.65	-	-	
	GLCM	88.51			
	DCI	73.56			
	DFT	70.69			
Jain et a. [28]	CNN	91.0	-	-	-
Baweja et al. [27]	CNN + Inseption-v3	93.0	-	-	-
Proposed Method	GLCM + AlexNet	**96.6**	**99.7**	**95.8**	**0.977**

Baweja et al. [27] who applied CNN method over leprosy images to perform leprosy prediction. Despite the effort to

improve CNN with Google inception-v3 model for leprosy region segmentations and learning, the highest accuracy observed was 93%, which is significantly lower than our proposed method (96.6%). A similar effort was made by Jain et al. [28] who applied CNN towards leprosy image classification. Unfortunately, with the classical pre-processing methods like median filtering, authors could merely achieve the average accuracy of 91% which is significantly lower than our proposed model (96.6%). Mondal et al. [29] too applied a modified deep learning architecture named Generative Adversarial Network (GAN) which was further improved with DensNet-121 could achieved the highest leprosy image classification accuracy of 94.25%. However, authors found that this claimed performance (i.e., the accuracy of 94.25%) is possible only with data augmentation. Interestingly, without data augmentation, the same GAN model exhibited accuracy of almost 83%, which is much lower than our proposed model. The comparison with other deep learning models like MobileNet V2 (81.6%), ResNet-101 (87.35%) and GoogleNet (89.65%) too are found lower than the proposed hybrid features driven solution. Lo et al. [30] too applied deep-CNN model (DCNN) in conjunction with AlexNet to perform bacillus image blocks classification. In addition to the fact that the images considered were not in sync with real-world diverse leprosy lesions and allied non-linear spatio-textural textures, the highest accuracy obtained was 73.8%. In this approach, authors applied a static threshold value of 0.5, where each image block was assessed for DCNN feature values. An image with probability value more than 0.5 was classified as bacillus. In sync with real-time non-linear and diverse leprosy lesions, this approach can't be guaranteed to exhibit reliable performance, especially over varying skin colours, non-linear shapes, sizes and orientation (due to the lesion presence on the different body parts). Noticeably, the proposed model addresses aforesaid real-time challenges and its functional efficacy confirms its robustness over the other existing approaches in terms of higher accuracy (96.6%), precision (99.7%), recall (95.8%) and F-score value (0.9771). F-score parameter which indicates the reliability of the classifier over non-linear and minimally annotated data condition too has been found sufficient (0.9771), thus guaranteeing the robustness of the proposed model towards touchless, automatic vision-based leprosy detection and classification system. Considering the contribution by Das et al. [16] who explored the different approaches including local binary patterns (LBP), GLCM, Discrete Cosine Transform (DCT), Discrete Fourier Transform (DFT) towards leprosy image classification found that the DCT-features can yield the highest accuracy of 73.56%, while the highest accuracy performance with GLCM was observed as 88.51%. In the same research, LBP was found exhibiting the leprosy image classification accuracy of 89.66%. DFT-based approach yields the least performance with the classification accuracy of 70.7%. The comparison results clearly indicate that the proposed hybrid feature driven model (i.e., *GLCM + AlexNet) outperforms other existing approaches or other feature driven concepts towards vision-based leprosy prediction.

Conclusion

This paper proposed a highly robust heuristic driven segmentation assisted hybrid deep-spatio-textural feature learning frameworkfor leprosy detection and classification. Unlike classical vision-based methods that extracts complete features from each input image, irrespective of the ROI-specific regions, this work focused on segmenting the ROI-specific regions for feature extraction. In other words, this study hypothesized that learning the features specific to the ROI-regions can help improving the accuracy of the leprosy lesion classification, which is often found very complex and non-linear in two-dimensional space. To enable automatic and improved ROI segmentation, in this paper a heuristic driven clustering method was developed that not only enabled precise ROI segmentation but also alleviated any need of manual seed-point definition. More specifically, in this work FFCM was developed to segment only leprosy specific regions (say, ROI). To guarantee optimal spatio-textural feature learning, this work retrieved RGB color space for the detected ROI by performing pixel-wise multiplication (i.e., multiplying segmented ROI regions with corresponding ROI-specific RBG pixels). Thus, obtaining ROI-specific RGB lesion space, the proposed model executed a hybrid feature extraction method encompassing GLCM and AlexNet deep learning algorithm. Here, to retain information rich feature, GLCM model extracted descriptive spatio-textural statistics encompassing entropy, energy, correlation, homogeneity, mean, standard deviation and variance information. On the other hand, to extract maximum possible deep features, transferable deep learning model AlexNet was applied, which obtained 4096-dimensional features to perform further learning and lesion-classification. Unlike classical standalone feature driven methods, (i.e., the one employing either GLCM or deep features individually), this work fused both GLCM and AlexNet features to improve learning and allied classification accuracy. In sync with the heterogenous feature set (encompassing both deep features as well as spatio-textural features) random forest classifier was applied to perform two-class classification. Being ensemble learning model random forest algorithm classified each input image as the normal image or the leprosy lesion. The results confirmed that the proposed method achieves superior accuracy of 96.6%, precision = 99.7%,

recall = 95.8%, F-score = 0.977 in comparison to the other state-of-art methods. Interestingly, this study revealed that the inclusion of hybrid features encompassing both GLCM and AlexNet enabled superior performance in comparison to the GLCM and CNN-driven methods individually. It shows that the amalgamation of spatio-textural statistics like GLCM and high-dimensional AlexNet can yield superior performance towards complex CAD tasks like leprosy detection and classification. On the other hand, the use of FFCM strengthened the proposed model to enable automatic and swift leprosy region segmentation. Thus, the strategic amalgamation of FFCM and hybrid features (GLCM and AlexNet) processed over ensemble random forest algorithm enables a robust, automatic and touchless vision-based CAD system for leprosy detection and classification.

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Declarations

Conflict of interest No conflict of interest.

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